Outliers and the Shape of Data

Name _____ Date _____ Calculator allowed Version 1

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KEY WORDS outlier *(tau tawhiti)* · mean *(toharite)* · median *(tau waenga)* · skewed

A Spotting outliers ●○○

Remember: An outlier is a value that is much larger or much smaller than the rest of the data.

1. Write the outlier in each set of data.

a) 12, 15, 14, 13, 48, 16 _____

b) 7, 8, 6, 9, 7, 1 _____

c) 150, 154, 149, 152, 98, 151, 155 _____

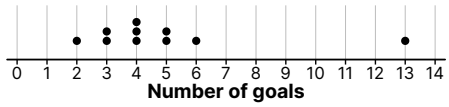
d) 33, 35, 30, 34, 31, 5, 32 _____

e) 60, 64, 61, 63, 62, 120 _____

f) 21, 2, 24, 22, 25, 23 _____

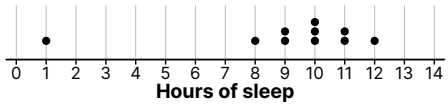
2. Write the outlier shown on each dot plot.

a)



_____

b)



_____

B Outliers, the mean and the median ●●○

Watch out: An outlier is not always a mistake. Check where it came from before you leave it out.

3. The ages, in years, of the people at a birthday party are: 11, 12, 12, 13, 12, 11, 13, 44

	Mean	Median
With all the values		
Without the outlier		

a) Which value is the outlier?

b) Complete the table. Round the means to 1 decimal place.

c) Which changes more when the outlier is left out, the mean or the median?

4. A height of 15 cm is written in a list of the heights of Year 7 students. Suggest how this outlier could have happened.

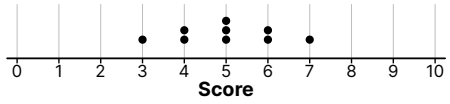
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C The shape of a distribution ●●○

Remember: Symmetrical data looks about the same on both sides of its peak. Skewed data has a longer tail on one side: skewed to the right has its tail on the right.

5. Describe the shape of each dot plot: symmetrical, skewed to the right or skewed to the left.

a)



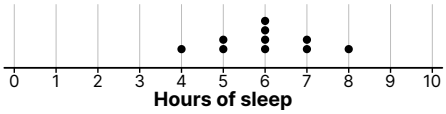
b)



c)



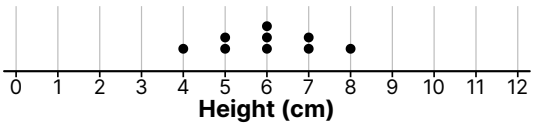
d)



D Shape, centre and spread ●●●

Remember: To describe a set of data, give its shape, its centre (the median) and its spread (the range).

6. The dot plot shows the heights, in centimetres, of the seedlings in a tray.

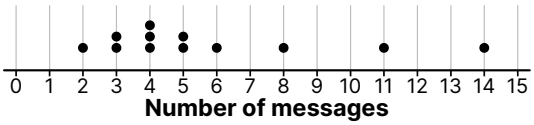


a) Describe the shape of the data.

b) Find the median (the centre).

c) Find the range (the spread).

7. The dot plot shows the number of text messages sent by each student in a group in one day.



a) Describe the shape of the data.

b) Find the median (the centre).

c) Find the range (the spread).

★ Extension

8. Six friends each have between $5 and $10 in their pockets. A seventh friend has $100. Would the mean or the median give a better idea of how much a typical friend has? Explain.

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Outliers and the Shape of Data

Name _____ Date _____ Calculator allowed Version 2

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KEY WORDS outlier *(tau tawhiti)* · mean *(toharite)* · median *(tau waenga)* · skewed

A Spotting outliers ●○○

Remember: An outlier is a value that is much larger or much smaller than the rest of the data.

1. Write the outlier in each set of data.

a) 11, 14, 13, 15, 52, 12 _____

b) 8, 9, 7, 9, 8, 2 _____

c) 160, 164, 159, 162, 105, 161, 165 _____

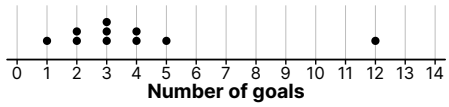
d) 43, 45, 40, 44, 41, 9, 42 _____

e) 70, 74, 71, 73, 72, 140 _____

f) 31, 3, 34, 32, 35, 33 _____

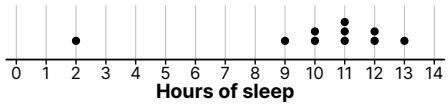
2. Write the outlier shown on each dot plot.

a)



_____

b)



_____

B Outliers, the mean and the median ●●○

Watch out: An outlier is not always a mistake. Check where it came from before you leave it out.

3. The ages, in years, of the people at a birthday party are: 12, 13, 13, 14, 13, 12, 14, 53

	Mean	Median
With all the values		
Without the outlier		

a) Which value is the outlier?

b) Complete the table. Round the means to 1 decimal place.

c) Which changes more when the outlier is left out, the mean or the median?

4. A height of 15 cm is written in a list of the heights of Year 7 students. Suggest how this outlier could have happened.

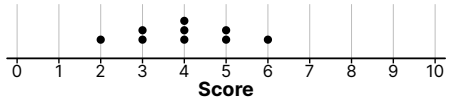
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C The shape of a distribution ●●○

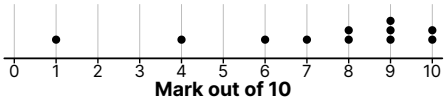
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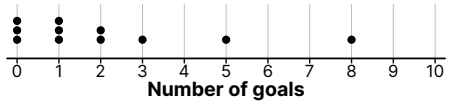
a)



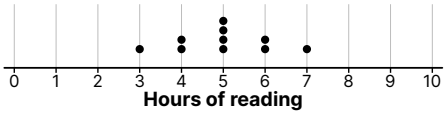
b)



c)



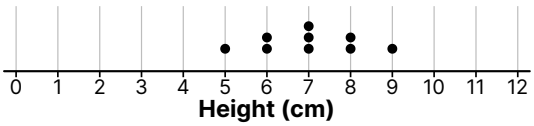
d)



D Shape, centre and spread ●●●

> **Remember:** To describe a set of data, give its shape, its centre (the median) and its spread (the range).

6. The dot plot shows the heights, in centimetres, of the seedlings in a tray.

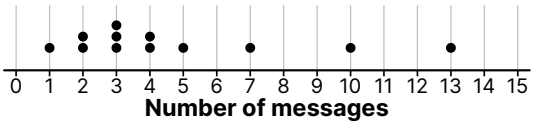


a) Describe the shape of the data.

b) Find the median (the centre).

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7. The dot plot shows the number of text messages sent by each student in a group in one day.



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